

Meeting Note 09

Date: 2026-02-12 **Time:** 14:00 - 14:30 **Location:** Supervision Office / Online **Attendees:** LIN Zhanzhi, CAO Yixin (Supervisor)

1. Objectives

- Review the implementation and results of the "Hybrid Parallelism" (Depth-Based Cutoff).
- Discuss the scaling breakthrough (breaking the 4.4x plateau).
- Address the performance regression observed in structured data patterns.

2. Progress Report

- **Hybrid Parallelism Implemented:**
 - Added a recursion depth check in `include/dpqs/parallel/parallel_sort.hpp`.
 - **Logic:** If depth > 20, the thread switches immediately to `sort_sequential` for the entire subtree, bypassing the scheduler.
 - **Tuning:** Reduced `MIN_PARALLEL_SORT_SIZE` from 64k to 8k to improve initial load balancing.
- **Benchmark Breakthrough (10M Integers):**
 - **16 Threads: 4.51x - 4.66x** Speedup (Up from 4.4x).
 - Successfully squeezed more performance out of the saturated system by removing scheduler overhead for leaf tasks.
- **Challenges:**
 - Regressions observed in `REVERSE_SORTED` (0.57x) and `ORGAN_PIPE` (0.76x) patterns compared to the ultra-fast sequential baseline.

3. Discussion & Feedback

Key Discussion Points

- **Sequential Fallback Success:** The improvement confirms that the overhead of creating/locking tasks was a significant factor at 16 threads. By removing it for the deep leaves, we reduced bus contention.
- **Structured Data Regression:** The supervisor pointed out that `REVERSE_SORTED` is handled in $\$O(N)\$$ by the sequential Dual-Pivot (it just reverses the run). Parallelizing it forces it into $\$O(N \log N)\$$ splitting, which is inherently slower. This is an expected trade-off.

Supervisor Feedback

"The 5% gain at 16 threads is significant because you are fighting against Amdahl's Law and physical hardware limits. It validates the Hybrid approach." Regarding the regression: "Do not view the performance drop on 'Reverse Sorted' as a failure. It is a known property of parallelizers. However, you should document it clearly in the report."

4. Issues & Challenges

- **No Further Scaling:** It is unlikely we can push beyond ~4.6x on this specific hardware without a complete algorithm rewrite (e.g., changing the partitioning scheme itself to be branchless or SIMD-

optimized), which is out of scope.

5. Action Items

- ☒ Implement Hybrid Parallelism.
- ☒ Benchmarking & Analysis.
- ☐ **Final Report Writing:** Begin merging the individual implementation reports into the Final Thesis structure.
- ☐ **Code Cleanup:** Ensure all debug comments are removed and headers are standardized.

6. Next Meeting Plan

- **Target Date:** 2026-02-19
- **Focus:** Final Report Skeleton Review.